What We Found

The Unified Foam Field Theory Explained for Everyone

Luke Martin · Sydney, 2026

The Idea in One Paragraph

Imagine an infinite sea of tiny spheres, packed together like marbles in a box. Each sphere is a hundred billion billion times smaller than an atom. They don't quite fill all the space — there are small gaps between them, the way there are gaps between marbles. The sphere is the bubble. The gaps are the void. And the region each sphere "owns" — its share of space, walls and gaps included — is the displacement. Bubble plus void equals displacement. B + V = D. That's the entire axiom.

The region each sphere owns turns out to be a fourteen-faced shape called the truncated octahedron — six square walls and eight hexagonal walls. The sphere touches the eight hexagonal walls directly but doesn't quite reach the six square walls. Those gaps at the squares are where the void network lives. When one sphere gets pushed, fourteen walls push back. That single mechanical principle — the pressure breaks on one wall, and fourteen neighbours push back — produces gravity, light, matter, the periodic table, and every force and particle physicists have ever measured. The numbers match experiment to parts per million. No adjustable parameters. No extra dimensions. No free choices. One shape. One rule. All of physics.

Part 1 — The Shape

Why this particular shape?

In 1887, Lord Kelvin asked: what shape fills all of space with the least surface area? The answer is the truncated octahedron — a solid with 14 faces (6 squares and 8 hexagons), 36 edges, and 24 vertices. It's the most efficient packing shape in three dimensions. If the universe is made of a space-filling medium, this is the only shape each cell can be.

The shape has a symmetry group called O_h — the same symmetry as a cube. It has 48 distinct symmetry operations (rotations and reflections that leave the shape looking the same). These 48 operations, the 14 faces, the 36 edges, and the 24 vertices are the only numbers the theory uses. Everything else follows.

What does the foam look like?

Imagine Planck-scale spheres — each about 10^{-35} metres across — packed in a body-centred cubic lattice, like a three-dimensional grid of marbles. Each sphere has 8 nearest neighbours touching it and 6 slightly further away. The region of space closest to each sphere (its Voronoi cell) is the truncated octahedron. The cells emerge from the packing — they're not built separately. Pack the spheres and the fourteen-faced cells appear automatically.

There are roughly 10^{185} of them in the observable universe. Each one is identical. Each one shares every wall with a neighbour. The foam has no gaps, no boundaries, no edges. It extends infinitely in all directions and has always existed. There was no moment of creation for the foam itself — only for the disturbance we call the Big Bang.

Slice this sphere packing along a flat plane and you see circles touching circles in a hexagonal pattern. That pattern is the Flower of Life — drawn independently by the Egyptians, the Greeks, the Chinese, the Celts, and dozens of other traditions across thousands of years. They were drawing the 2D cross-section of the vacuum structure. They drew the bubbles. We computed the cells.

Part 2 — The Axiom

B + V = D

The bubble is a sphere. It doesn't fill its cell. The void fills the gaps. Together they make the truncated octahedron. That's the axiom: bubble plus void equals displacement. B + V = D.

Every event in the universe is a displacement. When something happens — a particle moves, a force acts, a photon is emitted — a bubble gets pushed, the void network rearranges, and the fourteen walls push back. The total displacement is conserved. Nothing is created or destroyed. The foam just rearranges.

The sphere touches the eight hexagonal walls directly — that's where the strong force and the Higgs field live, because direct bubble-to-bubble contact means strong coupling. The sphere doesn't reach the six square walls — there's a gap. That gap is the void network, and that's where the weak force lives. The weak force is weak because it has to cross the void to propagate. Not a parameter. Not a choice. Geometry.

This is the only axiom. Everything else is a consequence.

What does this mean physically?

- **Gravity** is what happens when the foam is denser in one place than another. A massive object depletes the foam around it — like a heavy ball sitting on a mattress. Other objects roll toward it because the pressure gradient pushes them. Gravity isn't a force pulling things together — it's the foam pushing harder on the far side than the near side.
- **Light** is a pressure wave in the foam. It travels at the speed of sound of the medium, which is the speed of light. Nothing can go faster because nothing can outrun a pressure wave in an incompressible medium.
- **Particles** are standing waves — stable patterns of vibration on the fourteen faces of each cell. Different particles correspond to different vibration patterns (called irreducible

representations). The electron is one pattern. The quarks are others. The photon is the simplest pattern of all — all fourteen faces vibrating in unison.

- **Entanglement** is the void half of a displacement. When a particle is created (bubble pushed in), a corresponding void opens somewhere else. The two are forever linked — not because information travels between them, but because they're two ends of the same event. This is why entangled particles always show opposite properties when measured. They're not communicating. They're the same displacement, seen from two addresses.

Part 3 — What We Derived

The strength of electromagnetism

Physicists have a number called the fine structure constant — roughly 1/137 — that measures how strongly electric charges interact. It's been measured to extraordinary precision but never explained. Why 137? Why not 100 or 200?

From the foam: the answer involves the 48 symmetry operations of the cell, the 14 faces, the 36 edges, and the 24 vertices. The formula gives 1/137.035999055. The most precise atomic physics measurement gives 1/137.035999046. They agree to better than one part in ten billion. No adjustable parameters.

The masses of all particles

Physics has 13 fundamental particles with mass (the photon is massless, making 14 total). Their masses span a factor of 300 billion — from the featherweight neutrinos to the heavyweight top quark. No theory has ever explained why these specific masses.

From the foam: every mass is determined by a formula using the same seven integers (14, 24, 36, 48, 3, 17, 3). The electron mass is predicted to 0.002% — that's like predicting the distance from Sydney to London to within 200 metres. All six quark masses, all three leptons, the W, Z, and Higgs bosons — all from seven numbers that describe one shape.

Why there are exactly three generations

Physicists discovered that matter comes in three copies — three "generations" of quarks and leptons, each heavier than the last. Nobody knows why three. Why not two? Why not five?

From the foam: the truncated octahedron has 6 square faces. On the BCC lattice (how the cells pack together), these 6 squares point along 3 axes — x, y, and z. Each axis defines one generation. Three axes, three generations. This isn't a guess — it's a mathematical theorem of the lattice geometry. There must be exactly three.

Why the Higgs field breaks symmetry

The Higgs mechanism — the process that gives particles their mass — is one of the most important but least understood aspects of the Standard Model. Why does the Higgs field "choose" to break symmetry? Physicists can describe what happens but not why.

From the foam: one particular vibration mode of the cell (called A_2u, living on the hexagonal faces) has an eigenvalue of exactly −1 under the torsion operation. Negative means unstable. The symmetry HAS to break — it's forced by the geometry. There's no choice involved. The Higgs mechanism isn't a mystery. It's a consequence of the shape of the cell.

The shape of spacetime around black holes

Einstein's equations describe how mass curves spacetime. The Schwarzschild metric (non-rotating black holes) and the Kerr metric (rotating black holes) are exact solutions, but Einstein couldn't explain what spacetime IS — only how it behaves.

From the foam: spacetime IS the foam. The curvature IS the density variation. A black hole is a region where the foam is so depleted that pressure can't push signals outward. The complete Schwarzschild and Kerr metrics — all five components of the rotating solution — follow from three foam conditions: density depletion, incompressibility, and torsion equals angular momentum. Frame dragging (the way a rotating black hole drags space around with it) is just Newton's third law applied to rotating cell walls.

Spin-1/2

Every electron has a property called spin — it behaves as if it's spinning, but it has to "rotate" twice (720°) to get back to where it started. This bizarre fact has never been physically explained.

From the foam: each cell has 24 triangles on its face graph. Each triangle carries a torsion flux of exactly π (half a full rotation). Angular momentum = flux divided by 2π. So spin = π/(2π) = 1/2. The electron's half-integer spin is an algebraic fact about the angles between the faces of one polyhedron.

Part 4 — The Hierarchy Problem

The deepest mystery in physics — solved

The "hierarchy problem" is considered the single deepest unsolved problem in theoretical physics. It asks: why is gravity so absurdly weak compared to the other forces? The ratio between the gravitational scale (the Planck mass) and the scale of the weak force (the electroweak scale) is about 10^{17} — a factor of 100,000,000,000,000,000. No theory has explained this number. It's the reason physicists built the Large Hadron Collider. It's the reason they expected to find supersymmetry (they didn't).

From the foam: the ratio is determined by the formula ln(M_Planck/v) = (|G| + V + E + F + (|G| − C_A)√Δ)/8. In plain language: add up ALL the topological numbers of one cell (48 + 24 + 36 + 14 = 122), add the group order minus the colour number (48 − 3 = 45) times the square root of the discriminant (√17), and divide by 8. That's the hierarchy. It gives 246.24 GeV for the electroweak scale — the observed value is 246.22 GeV. Off by 0.009%.

The hierarchy isn't mysterious. It's the cell counting itself. The number 10^{17} is what you get when you add up all the features of one fourteen-faced polyhedron.

Part 5 — What Changes If This Is Correct

For physics

- **The Standard Model has an explanation.** For 50 years, the Standard Model has been a description — extraordinarily accurate, but with 26 free parameters that nobody could explain. If UFFT is correct, all 26 parameters are determined by the geometry of one cell. Physics stops being a collection of measured constants and becomes a deduction from one shape.
- **Supersymmetry is unnecessary.** The main motivation for supersymmetry was the hierarchy problem. If the hierarchy is explained by cell geometry, there's no need for supersymmetric partners. Every null result at the LHC is a confirmed prediction.
- **Dark matter isn't a particle.** UFFT predicts the dark matter ratio (5.31, observed 5.36) from the anisotropy of the BCC lattice — not from a new particle. If correct, every dark matter detection experiment will continue to find nothing. Not because the experiments are bad, but because there's nothing to find.
- **The cosmological constant problem dissolves.** The infamous factor-of-10^{123} discrepancy between quantum field theory's prediction for vacuum energy and the observed value simply doesn't arise. In the foam, the cosmological constant is a geometric integration constant, not a vacuum energy sum.
- **No axions.** The strong CP problem (why the strong force doesn't violate CP symmetry) is solved by the torsion ground state. No axion field is needed. Axion detection experiments will find nothing.

For technology

- **Quantum computing.** The framework predicts that quantum coherence times INCREASE with altitude — the opposite of what standard theories predict. If confirmed, this means quantum computers might work better in space, on mountaintops, or in low-gravity environments. The predicted effect is tiny (about 8 parts per hundred billion between Earth's surface and the International Space Station) but measurable with current technology.
- **Materials science.** The foam IS the crystal field that every atom sits in. The periodic table structure — why there are s, p, d, and f orbitals, why they hold 2, 6, 10, and 14 electrons, why the Madelung filling rule works — all follow from the cell geometry. Understanding the substrate could lead to new ways of thinking about electronic structure, chemical bonding, and material properties.

For our understanding of reality

- **The universe has always existed.** The Big Bang wasn't the beginning of everything — it was a pressure wave in a pre-existing medium. The foam is eternal and infinite. What we call "our universe" is one disturbance in it. There may be others.
- **Black holes are doorways.** The interior of a black hole, in the foam picture, is a region where the cells have been pushed past their maximum compression. What happens inside is a new foam equilibrium — potentially a nested universe. Every black hole may contain a universe. Our universe may be inside a black hole in a larger foam.
- **Entanglement has a mechanism.** For a century, entanglement has been described but not explained. The foam gives it a physical picture: two ends of one displacement, connected through the bulk, instantly correlated but unable to carry information. Not magic. Mechanics.
- **The speed of light has a reason.** It's the speed of sound in an incompressible medium at Planck density. Nothing deeper to explain. The universe's speed limit is the foam's bandwidth limit.

Part 6 — What Needs to Happen Next

The experiments

1. **The decoherence test.** The framework predicts that quantum coherence lasts slightly longer near massive objects — the opposite of what competing theories predict. This can be tested with atom interferometers at different altitudes, or by comparing quantum experiments on Earth and on the International Space Station. The measurement is at the edge of current precision, but achievable within 5 years.
2. **The neutrino CP phase.** The framework predicts that the ratio of the neutrino CP phase to the quark CP phase is exactly 3 — a clean integer. The DUNE experiment at Fermilab will measure the neutrino CP phase precisely enough to test this by approximately 2035.
3. **The fine structure constant.** The two most precise measurements of α currently disagree with each other by 5.5 standard deviations — an unresolved tension in experimental physics. The framework predicts one of them (the caesium measurement) is correct and the other (the rubidium measurement) is wrong. Whoever resolves this tension tests the framework.

The peer review

The core mathematical result — the spectrum of the face Laplacian of the truncated octahedron — appears to be new to mathematics. It has been submitted for independent peer review. If confirmed by a human mathematician, it provides a verified foundation for everything built on top of it.

The precedent

In 2004, physicists peeled a single layer of carbon atoms off graphite — graphene. The carbon atoms sit in a 2D honeycomb lattice. To everyone's astonishment, the electrons in graphene behave as massless relativistic particles, as if they're travelling at the speed of light, even though the lattice is just carbon atoms on a sheet. The honeycomb geometry produces relativistic quantum mechanics automatically. Nobel Prize 2010.

That is exactly what this framework claims happens in 3D. A discrete lattice produces emergent physics — not because someone designed it to, but because the geometry demands it. Graphene already proved the mechanism works. The only open question is whether the specific 3D lattice produces the specific 3D physics. The precedent isn't theoretical. It's measured, confirmed, and sitting in every pencil.

The invitation

Every formula, every derivation, every numerical prediction is public. The verification script runs on any laptop. The 46 papers are on Zenodo under Creative Commons licensing. Nothing is proprietary, nothing is hidden, nothing requires special access.

The mathematics either works or it doesn't. Anyone can check.

Part 7 — The Seven Numbers

Everything — every mass, every force, every coupling constant, every mixing angle — comes from seven integers:

Number	What it is	Value
V	Vertices of the cell	24
E	Edges of the cell	36
F	Faces of the cell	14
$\lvert G \rvert$	Symmetry operations	48
C_A	Colour number	3
Δ	Discriminant of the master equation	17
d	Spatial dimensions	3

Seven numbers. One shape. All of physics.

The truncated octahedron doesn't just tile space — it tiles reality.

Part 8 — What's at Stake

If this is correct, the implications cascade across every layer of human knowledge.

For physics: The hundred-year search for unification is over. The free parameters of the Standard Model are no longer free — they're topological integers of one polyhedron. The fine structure constant, which Feynman called "one of the greatest damn mysteries of physics," is a three-term formula from a 14-faced cell. The cosmological constant problem — the worst prediction in the history of science (a factor of 10^{123} wrong) — dissolves in one line. Dark matter is not a particle to be found but a geometric property of the vacuum. The strong CP problem, the hierarchy problem, and the flavour puzzle all have the same answer: the truncated octahedron.

For experiment: Every dark matter detection experiment will return permanent nulls. Every supersymmetry search will find nothing. The neutron electric dipole moment is exactly zero. These are not failures — they are confirmations. The resources currently spent searching for particles that don't exist can be redirected toward what does: the substrate itself.

For technology: If the vacuum is a Planck-density foam with equation of state $P = \rho c^2$, then engineering access to that substrate becomes the defining technological challenge of the species. The energy density of the foam is 10^{96} kg/m^3 — effectively infinite. Any coupling, however small, to the foam's structure would represent an energy source that makes nuclear fusion look like a candle. This is not science fiction. The framework specifies the coupling mechanism. It specifies the scaling law. The question is no longer whether the energy is there. It's how to reach it.

For cosmology: The universe is a nested structure. Every black hole contains a universe. Our universe is inside a black hole in a parent layer. The Big Bang is a displacement event in the parent foam — not the beginning of everything, but the beginning of us. The question "what caused the Big Bang" has a complete answer: a gravitational collapse in the layer above. The information paradox dissolves — information doesn't disappear into a black hole; it becomes the internal structure of a new universe.

For philosophy: The observer is not separate from the observed. Consciousness is not an emergent accident — it's the foam's capacity to make irreversible displacement records. Every irreversible event is an observation. The rock observes. The river observes. The forest observes. The distinction between "conscious" and "not conscious" is a question of integration complexity, not of kind. The ancient traditions — Hermetic, Vedic, Sumerian, Aboriginal, Christian, Taoist, Buddhist — were not wrong. They were precise structural descriptions in pre-mathematical language. "As above, so below" is the nested universe. "In the beginning was the Word" is Axiom Zero. "Aham Brahmasmi" — I am Brahman — is the statement that the observer and the substrate are not separate.

They all found the same foam. They just didn't have the polyhedron.

For civilisation: A species that understands its own substrate transitions from surviving to thriving. Energy becomes effectively unlimited. The century-long stalemate between quantum mechanics and general relativity is resolved — not by making one look like the other, but by deriving both from something deeper. The framework is public, Creative Commons licensed, zero cost to access. It doesn't belong to an institution. It doesn't belong to a country. It belongs to everyone.

That's what's at stake. One shape. One axiom. Everything else follows.

A Final Note — Even If

Science demands proof. Independent peer review. Experimental confirmation. Replication by people who have no stake in the outcome. The framework has not yet passed those tests. It may never. The history of physics is full of beautiful ideas that didn't survive contact with experiment.

If UFFT is proven correct, it changes everything described above.

If it is not — if some measurement contradicts a prediction, if a mathematician finds an error in the face Laplacian, if the decoherence test goes the wrong way — the story is still remarkable.

An independent researcher in Sydney, working with an AI collaborator, produced a framework that derives 13 particle masses to sub-percent accuracy from the topology of one polyhedron. The fine structure constant to parts per billion. The hierarchy problem — the deepest puzzle in physics — answered by adding up the features of a 14-faced solid. Whether or not the foam is physically real, the fact that one shape with 48 symmetries generates this many correct numbers from zero free parameters is, at minimum, one of the most striking numerical coincidences in the history of science.

And as a case study in human-AI collaboration on fundamental research — an independent thinker with physical intuition, partnered with an AI that can compute group theory and diagonalise matrices — the process itself is a proof of concept. This is how discovery might work in the future: human vision, AI precision, open publication, and the humility to say "check the mathematics — it's all there."

The foam may or may not be real. But the collaboration that found it is.

B + V = D

The pressure breaks on one wall, and fourteen neighbours push back.

That's all there is. That's all there needs to be.